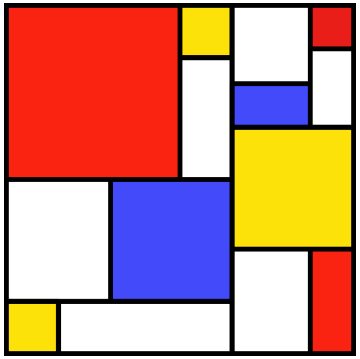
SLiM

Workshop Series



Callbacks III (`modifyChild()`)

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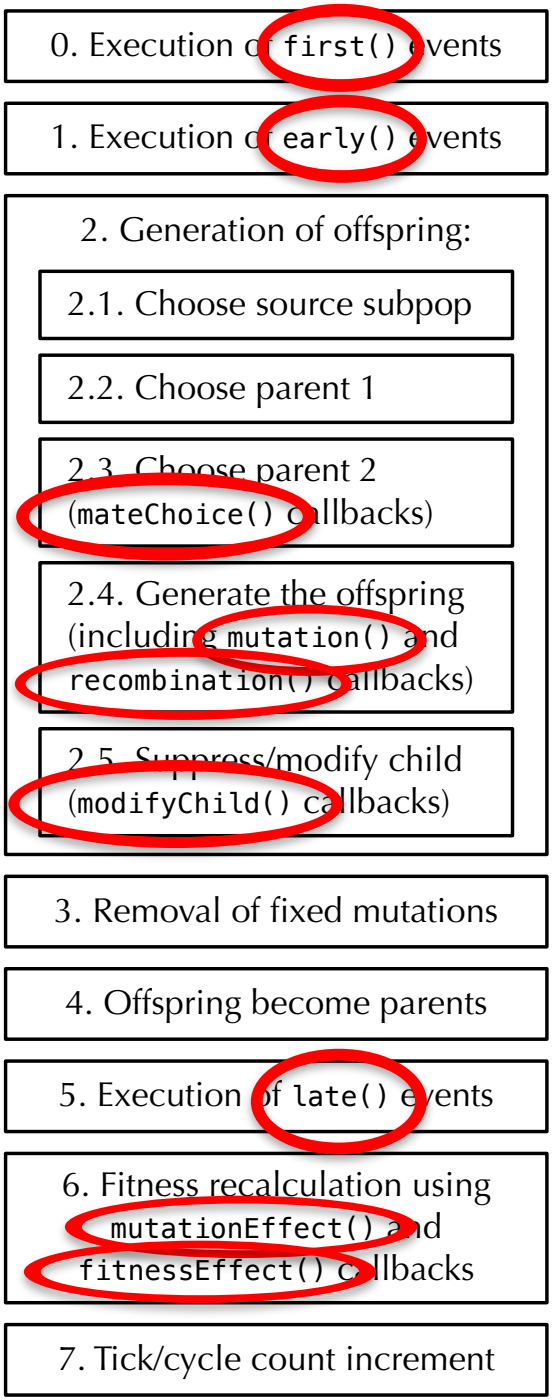
The WF model's tick cycle



The WF Tick Cycle

- Callbacks:
  - `first()` and `early()` events
  - `mateChoice()` callbacks
  - `mutation()` callbacks
  - `recombination()` callbacks
  - `modifyChild()` callbacks
  - `late()` events
  - `mutationEffect()` callbacks
  - `fitnessEffect()` callbacks
  - `interaction()`, `survival()`, `reproduction()`, ...

The WF model's tick cycle

0. Execution of `first()` events

1. Execution of `early()` events

2. Generation of offspring:
   - 2.1. Choose source subpop
   - 2.2. Choose parent 1
   - 2.3. Choose parent 2 (`mateChoice()` callbacks)
   - 2.4. Generate the offspring (including `mutation()` and `recombination()` callbacks)
   - 2.5. Suppress/modify child (`modifyChild()` callbacks)

3. Removal of fixed mutations

4. Offspring become parents

5. Execution of `late()` events

6. Fitness recalculation using `mutationEffect()` and `fitnessEffect()` callbacks

7. Tick/cycle count increment

`modifyChild()` callbacks

- called in offspring generation
- override default generation
- modify generated offspring
- often used to implement:
  - lethal mutations / epistasis
  - spatial dispersal / boundaries
  - gene drive mechanics
  - self-incompatibility
  - non-genetic inheritance

The WF model's tick cycle

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6. Fitness recalculation using `mutationEffect()` and `fitnessEffect()` callbacks
7. Tick/cycle count increment

`modifyChild()` callbacks

```
modifyChild([<subpopID>])
{
    ...
}
```

- apply to *one or all* subpops
- called for each **focal offspring**
- provided with that offspring
- return value:
  - **T** (proceed with generation)
  - **F** (terminate generation, draw new parents)

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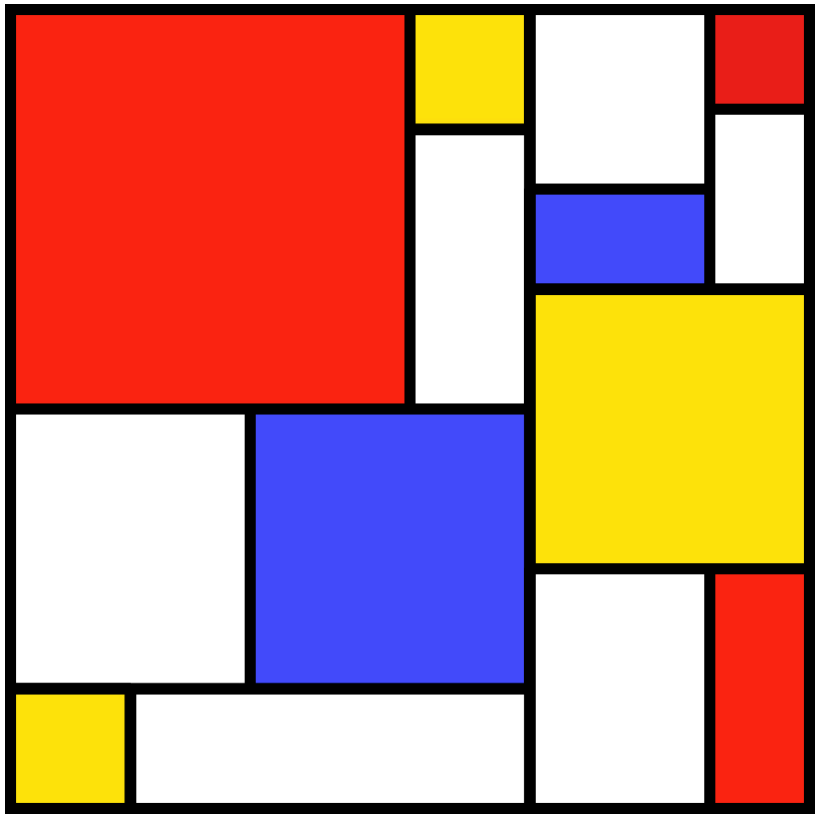
`modifyChild()` callbacks

- there is a *focal offspring*
- **pseudo-parameters:**
  - `child`
  - `parent1`
  - `parent2`
  - `isCloning`
  - `isSelfing`
  - `subpop`
  - `sourceSubpop`



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